Chapter 6
Political Elites and Polarization: Understanding Political Discourse in the European Parliament During Turbulent Times

Hilke Brockmann, Wiebke Drews, Pedro Fierro, Joris Frese, and Daniel Triana

Abstract In liberal democracies, both experts and laypeople are observing an increase in polarized public discussions. The rise of social networks and various crises are linked to this political fragmentation. What role do political elites play in this? Our study examines European elite politicians at the start of the Russia-Ukraine war. Using a dataset from Twitter/X and an unsupervised, pre-trained, transformer-based topic model, we analyze the discourse of all members of the European Parliament (MEP) and the European Commission. Our findings show that the attention of political elites to the conflict is immediate and brief. Sentiment analysis confirms the polarized nature of discussions related to the crisis. Notably, male politicians and those on the ideological extremes tend to express more negative sentiments than females and politicians of the political center. Additionally, a crisis event like the outbreak of the Russian-Ukrainian war tends to deepen this emotional divide. Our research adds to existing literature on polarized narratives by focusing on polarization dynamics driven by diverse political elites.

Hilke Brockmann, Wiebke Drews, Pedro Fierro, Joris Frese and Daniel Triana authors contributed equally and are listed in alphabetical order.

H. Brockmann (✉)
Constructor University, Bremen, Germany
e-mail: hbrockmann@constructor.university

W. Drews
University of the Bundeswehr Munich, Munich, Germany

P. Fierro
University Adolfo Ibanez, Santiago, Chile

J. Frese
European University Institute, Florence, Italy

D. Triana
National Autonomous University of Mexico, Mexico City, Mexico

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6.1 Introduction

Democratic institutions are coming under attack worldwide. The Economist reports a decrease of 5% in the Democracy Index in Europe and 12% in North America since 2006 (Economist Intelligence Unit 2023). Growing polarization is one of the reasons why democratic compromise is at risk, some experts say (Haggard and Kaufman 2021; Kingzette et al. 2021). However, polarization is not a new phenomenon. The Oxford English Dictionary dates the political connotation of the term back to 1862 (OED 2023; Schedler 2023). Different definitions circulate in the literature. Most recent papers refer to clusters of growing ideological and affective distance between social groups or parties, but do not specify clear boundaries (Schedler 2023). Thus, some empirical studies show a recent rise in polarization (Finkel et al. 2020), while others do not (Kreiss and McGregor 2023; Mau et al. 2023).

At the same time, the large supply and demand for polarization research is often linked to the rise of populism (Guriev and Papaioannou 2022), and to the simultaneous spread of social media (Brady et al. 2021; Harris et al. 2023). Digital platforms provide a perfect breeding ground for new and divergent political deliberations (Lorenz-Spreen et al. 2022). They allow responses in real-time, at no cost, and without geographical or social boundaries. Moreover, recommender algorithms tend to automatically connect like-minded individuals (Cinelli et al. 2021). These features make social media valuable communication tools even for established politicians. Since politicians are now more active on social media than ever, and since social media seems prone to polarized debates (Allcott et al. 2020), we are interested in whether political elites contribute to polarized political discussions.

Specifically, we aim to analyze how politicians respond to external shocks and whether these shocks can lead elite-level political communication to become more negative and polarized.

On social media, we follow an entire political elite during the outbreak of the Russia-Ukraine war. In times of acute crisis, we expect certain politicians to engage in new and polarized discourses to mobilize support for their political positions. We anticipate polarization to manifest as increasingly negative sentiments and widening sentiment gaps between opposing political camps.

Our findings shed light on the dynamics of polarization. They also may help us better assess whether, during times of polycrisis (Tooze 2022)—when people lack orientation because there is no single cause and simple fix—political polarization is a more likely response from elite politicians than political restraint.

The organization of the paper is as follows: First, we review the literature on polarization, social media, and political elites. Then, we explain our methodological approach. Our analysis is based on more than half a million tweets from all members of the European Parliament and the European Commission during the four months

before and after the outbreak of the Russia-Ukraine war on February 24, 2022. We use machine learning methods—namely Transformer models and sentiment analysis—to identify topics and sentiments during the peak of the crisis. We present our results and conclude.

6.2 Literature Review

Political polarization, the growing ideological and emotional distancing between social groups (Phillips et al. 2023), is neither a new phenomenon nor a steady state. The concept encompasses both processes and outcomes (Arora et al. 2022). Its demand in scientific and public discussions is significant and growing however, a clear definition and consistent measurement are lacking (Kubin and Von Sikorski 2021). Still, there is a common understanding that political polarization refers to inter-group dynamics like social segregation and sorting, creates in-group identity and solidarity, and out-group differentiation or even demonization, and takes place in uncompromising, hardened political deliberations, which poses a threat to democracy (Haggard and Kaufman 2021; Wilson et al. 2020). Often unforeseen events trigger polarization processes (Macy et al. 2021; Mainych et al. 2023; Svozil et al. 2025; Weber et al. 2024; Westwood and Peterson 2022).

6.2.1 External Shocks and Crises

Politics is frequently driven by external shocks and crises. These extraordinary events offer elite politicians an opportunity to raise their profile. For example, during the COVID-19 pandemic polarization among U.S. Congress members grew stronger along the ideological lines of the two main political parties. Each party adopted different communication strategies that reflected and reinforced existing ideological divisions (Green et al. 2020). Similarly, economic recessions, which can be seen as a type of external shock, may lead to increased polarization as they frequently spark debates over government responsibility in their prevention, creation, and mitigation, as well as related market regulation and social welfare policies. These debates can deepen pre-existing ideological divisions (Funke et al. 2016; Macy et al. 2021).

However, shocks and crises not only exacerbate divisions among social groups, but they also serve as unifying forces that (temporarily) narrow social gaps by fostering a sense of group identity (Levendusky 2018). When a society or specific community faces wars, natural disasters, terrorist attacks, or global pandemics, the initial response can often trigger a mobilization effect, briefly reducing polarization as individuals and groups rally against a common threat. Events like the Russian invasion of Ukraine have been studied for their influence on cohesion or division among political elites. For instance, Chueri and Törnberg (2024) examined its influence on the European Parliament's unity through Twitter network analysis. While the crisis

initially seemed to enhance communication and unity among members, this effect was temporary and did not lead to lasting cohesion. Despite increased focus on the war and a surge in European solidarity, pre-existing political and geographical divisions within the European Parliament persisted, and Eurosceptic groups remained isolated. The study concludes that the invasion did not produce enduring unification among European political elites, leaving existing fractures intact (Chueri and Törnberg 2024).

In our study, we examine the Russian invasion of Ukraine on February 24, 2022, as an pivotal turning point in European politics, focusing on how various politicians discuss the war and related issues before and after the large-scale invasion. First, we expect that European politicians will immediately shift their focus to the Russia-Ukraine war (H1: Attention Hypothesis).

6.2.2 Political Polarization and Social Media

Whether social media is a decisive factor in the rise of polarization is controversial (Graham and Svolik 2020; Lorenz-Spreen et al. 2022; Sunstein 2018; Zhuravskaya et al. 2020), as the interplay between social media and polarization turns out to be complex. Extensive research (Bail et al. 2018; Bakshy et al. 2015) investigated if and how exposure to ideologically diverse content on platforms like X/Twitter and Facebook affects the political views of users. Most findings prove that exposure to social media will reinforce existing beliefs rather than promote understanding. This narrowing within one camp is further supported by the algorithms that govern content distribution on these platforms and amplify partisan content (Banks et al. 2021). The role of platform design and user interaction is also critical to drive groups apart. Research shows that social feedback mechanisms reinforce expressions of moral outrage and polarization (Brady et al. 2021; Harris et al. 2023).

Beyond ideological dividing lines, emotions shape the polarization of political discourse and opinions (Webster and Albertson 2022). Affective polarization is connected to partisanship and elicits positive feelings towards the in-party but, more importantly, negative emotions towards the out-party. Particularly in situations of political threat, negative emotions prevail (Druckman et al. 2020; Iyengar et al. 2019).

Anger, resentment, and dissatisfaction ignite political movements for social change, regardless of the politician's gender (Traister 2018). Mechkova and Wilson (2021) demonstrate that women who make use of an angry rhetoric on social media are more likely to win a vote than more reserved female candidates. But women who challenge gender stereotypes also attract more hate comments. Overall, women show more emotions than men, with female politicians expressing more positive emotions while male politicians show more negative affect (Brescoll and Uhlmann 2008; Renner and Masch 2019).

Content that expresses animosity towards out-groups has been found to drive engagement on social media platforms (Rathje et al. 2021), indicating that the

nature of the content itself, as well as the interactions it fosters, can contribute to polarization. This points to a potentially concerning aspect of social media: their tendency to reward content that fuels division and conflict, which can have real-world consequences for subjective well-being, social cohesion, democratic discourse, and political voting behavior (Allcott et al. 2020; Guriev and Papaioannou 2022).

In line with this research on polarization, we hypothesize that any political communication about the conflict will exhibit significantly more negative feelings than communication about other issues (H2a: Feeling Hypothesis). Men's conflict-related communication should be significantly more negative than that of women, as research has shown that women face harsher backlash when expressing negative emotions (H2b: Gendered Sentiment Hypothesis). In addition, ideologically extreme politicians are likely to express more negative emotions than moderate politicians because social media algorithms tend to amplify more engaging communication, and a marginalized political position requires fewer compromises in that regard (H2c: Ideological Sentiment Hypothesis). Lastly, we assume that politicians geographically close to Ukraine express more negative emotions than those representing countries further away, as negative emotions better emphasize the greater threat of the situation (H2d: Geographical Sentiment Hypothesis).

6.2.3 Political Elites

Elite polarization, particularly as it manifests within social media, has become a major topic of academic interest and social concern (Levin et al. 2021). Political divisions are wider, and extremist views are more common among elites than among ordinary voters (Macy et al. 2021). Theoretical perspectives often draw on the concept of echo chambers and filter bubbles (Pariser 2011; Sunstein 2001), where algorithmic curation strengthens existing beliefs and reduces exposure to opposing perspectives (Cinelli et al. 2021). This self-reinforcing cycle is thought to exacerbate ideological divides, as elites engage in performative polarization to galvanize their base, often at the expense of cross-ideological dialogue (Tucker et al. 2018). Furthermore, the architecture of social media platforms, which prioritizes engagement metrics such as likes, shares, and comments, may incentivize elites to adopt more radical positions to attract viewers (Bakshy et al. 2015). Therefore, we expect the emotional distance between ideological groups to grow after the Russian invasion of Ukraine (H3: Polarization Hypothesis).

6.3 Data and Methods

Through the Academic Twitter API,[1] we scraped every tweet posted by members of the European Parliament (MEPs) and the European Commission from October 13, 2021 to July 8, 2022. This period covers nineteen weeks before and after the start of the Russian invasion of Ukraine. In total, our dataset includes 520,871 tweets from over 600 parliamentarians and all commissioners at that time. Both original tweets and retweets are included. In addition to the data from the API, we enrich our dataset with a comprehensive set of demographic and political variables related to these EU politicians, such as their gender, age, country of origin, EU party group affiliation, national party affiliation, and the political position of their party (and their EU party family) on various policy issues (European Parliament 2022; Jolly et al. 2022; Kinkartz et al. 2019).

Since the tweets were originally posted in various languages, we first translated them into English, using the Python package deep-translator 1.1. (Baccouri 2020). The deep-translator 1.1 Python package is a versatile tool for machine translation, designed to facilitate translating texts from various languages into English. It connects to multiple translation services like Google Translate, Microsoft Translator, and others, providing broad language support. This package functions by sending the input text to the selected translation service's API, which then returns the translated version. When manually validating a random sample of translated tweets, the quality of the translations achieves face validity.

For the subsequent thematic and sentiment analyses, it was essential to preprocess the tweets: firstly, tweets were decoded from HTML and stripped of specific tags and URLs. The preprocessing also included normalizing quotation marks, unicode characters, and handling non-ASCII characters and whitespace. Additionally, Twitter handles, emails, and accents were removed, while dashes and special characters were replaced. The text was further converted to lowercase, with the removal of numbers and special characters, focusing solely on alphabetic characters. This meticulous cleaning ensured that the tweets were in a consistent and analyzable format.

In our thematic analysis, BERTopic is employed to extract latent topics from texts (Grootendorst 2022, p. 1). It starts by creating document embeddings, which are numerical representations that capture the semantic meaning of texts, using the all-mpnet-base-v2 sentence-transformers model.[2] This model tends to capture a wider range of semantic information and is often chosen for tasks that require a deep understanding of sentence context and structure. It is well-suited for processing short texts like tweets, because it is trained on over 1.17 billion sentences from various sources, including Reddit, WikiAnswers, and the S2ORC. After generating embeddings, BERTopic employs HDBSCAN, a clustering algorithm, to group similar embeddings and identify related topics. It then applies a class-based TF-IDF method to

[1] Data collection took place in mid-2022, i.e., before Elon Musk's takeover and the subsequent disablement of Twitter's Academic API.

[2] The sentence transformers model can be found here: ttps://huggingface.co/sentence-transformers/all-mpnet-base-v2.

select and emphasize key words for each topic cluster, helping to clarify the thematic content of each cluster. This structured approach allows for accurate extraction and interpretation of topics from our dataset, especially useful for short and informal texts such as tweets.

In our thematic analysis using BERTopic, we set the min_cluster_size parameter to 500 in the HDBSCAN clustering algorithm. This choice is crucial because it determines the minimum number of tweets required for a cluster to be considered a distinct topic. Choosing 500 as the minimum cluster size strikes a strategic balance. On the one hand, it ensures that each identified topic is supported by enough data, providing statistical significance and reliability. On the other hand, it prevents topics from fragmenting into too many small, potentially overlapping subtopics, which could complicate interpretation and dilute the thematic focus. This threshold is especially suitable for large datasets like ours, consisting of tweets, where it is important to identify substantial and coherent thematic clusters without missing meaningful patterns due to overly strict clustering criteria. A list of the top 20 topics classified by BERTopic is provided in Table 6.1.

To identify demarcated and polarized tweets, we classify emotional intensities in conversations. For this, we use the VADER (Valence Aware Dictionary for Sentiment Reasoning) lexicon (Hutto and Gilbert 2014). NLTK's VADER is a tool commonly

Table 6.1 Topics MEPs and the EU Commission tweeted about in 2021/22

Topics	Label	N (tweets)
1	Russia-Ukraine	55,991
2	Covid-19	35,847
3	France	30,125
4	Economy and finance	25,586
5	EP	19,639
6	Agriculture	18,662
7	Women's rights	17,610
8	Elections	16,304
9	Energy and resources	11,066
10	Immigration	10,940
11	Adjectives	9225
12	Gratitude	9142
13	Climate change	9079
14	Spain	8993
15	Digitization	8390
16	Italy	7484
17	Energy prices	7427
18	Poland	5963
19	MEP delegations	5505
20	Rule of law and executives	5147

used in NLP, designed to analyze sentiments in text. It is particularly effective at handling social media text, where sentiments are often expressed informally. VADER uses a combination of a sentiment lexicon and rules to assess the emotional tone of a text segment. It provides a compound score calculated by summing the valence scores of each word in the lexicon, adjusting them based on rules, and normalizing the results between −1 (most extreme negative) and + 1 (most extreme positive). One advantage of VADER over other sentiment analyzers like Lexicoder is its ability to incorporate nuances like sarcasm and context into the sentiment output.

6.4 Results

In the following subsections, we first show the distribution of topics over time and the prevalence of the Russia-Ukraine topic across different groups, including gender, party affiliation, and ideology (measured by the GALTAN scale). Then, we highlight divergences and polarization in expressed sentiments before and after the Russian invasion of Ukraine, and again across various groups, including gender, party affiliation, and political ideology.

6.5 BERTopic Model

The BERTopic analysis of the dataset yielded a total of 65 distinct topics. Within these, we identified and merged five closely related topics into one comprehensive theme centered around Russia-Ukraine. This combined topic covers a wide range of issues: general Russia-Ukraine communication, discussions about Russian energy supplies during the conflict, insights into Ukraine's EU and NATO accession status, attention to Ukrainian refugees, and specific discourse on the German-Ukraine relationship.

This amalgamation resulted in a significant topic cluster, accounting for a total of 55,991 tweets related to the Russia-Ukraine topic over the entire observation period. Consequently, the Russia-Ukraine topic is the second most frequently discussed topic in the dataset, surpassed only by a more general topic that expresses gratitude or announces events (for example, the most frequent terms in this topic are 'thank', 'today', and 'right'). Figure 6.1 visualizes the frequency of tweets addressing the ten most common topics over time, including the Russia-Ukraine topic. In the legend (termed 'global topic representation'), the most frequent words for each topic are displayed. The figure reveals notable patterns in the data. There was an abrupt surge in the volume of Russia-Ukraine-related tweets following the Russian invasion on February 24, 2022, which aligns with our attention hypothesis. In fact, attention toward this topic was already increasing in the weeks before the invasion, likely due to build-up and anticipation effects. However, the intensity of this communication rapidly declined in March. This pattern suggests that while an exogenous shock,

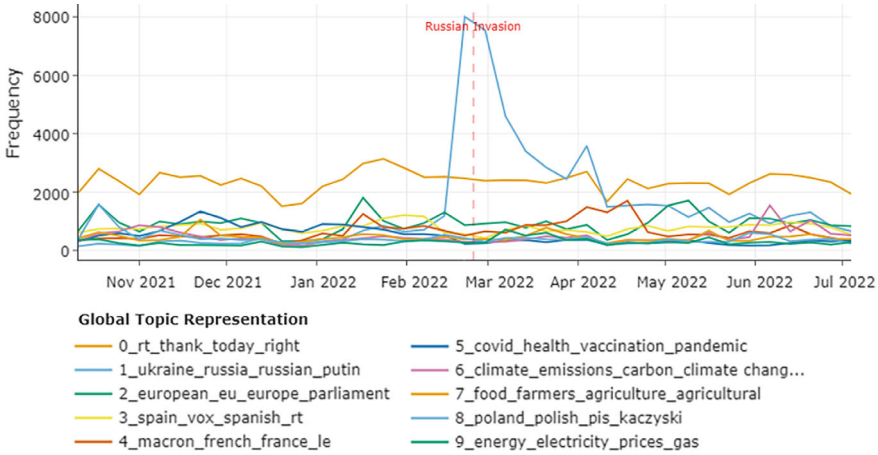

Fig. 6.1 Ten most frequent topics over time

such as the invasion, can trigger bursts of communication, interest among MEPs and Commissioners in the topic appears to wane shortly afterward. This observation might reflect the dynamic nature of political communication on social media, where initial reactions to major events are often intense but can quickly give way to other emerging issues.

To identify which EU politicians are most actively contributing to discussions about the war in Ukraine, we first visualize the percentage of Russia-Ukraine tweets out of all tweets from politicians in each EU country. As shown in the right panel of Fig. 6.2, where dark colors represent low percentages and light colors indicate high percentages, politicians from Eastern European countries discuss the Ukraine war in a much larger share of their tweets than those from Western Europe. For example, Polish politicians discuss the war in over 20% of their tweets, while Estonians do so in over 40%. In contrast, Spanish or French politicians mention it in less than 15% of their tweets. Since the total number of tweets varies significantly between countries, we also visualize the share of all war-related tweets for each country relative to the total across all countries, to better illustrate who dominates the discourse. In the left panel of Fig. 6.2, we see that Polish politicians contribute the largest share of war-related tweets compared to all other EU countries, with German politicians posting the second highest number. These findings suggest that geographical proximity greatly increases interest in the conflict.

After the invasion on February 24, the proportion of Russia-Ukraine related tweets increased across various groups, including gender, party affiliation, and ideology of EU politicans, as shown in Figs. 6.3, 6.4, and 6.5. However, some differences are noticeable: men discuss the topic significantly more than women (although it is debatable whether this difference is also substantially meaningful: roughly 18% of post-war male tweets are Russia-Ukraine related compared to about 16% of post-war female tweets). Furthermore, after the invasion, moderate politicians talk about the

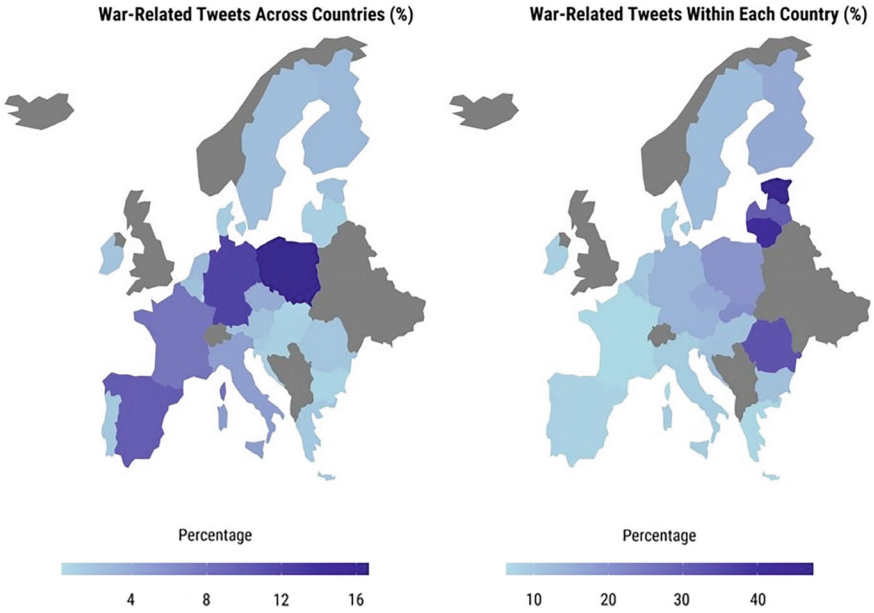

Fig. 6.2 Percentage of Russia-Ukraine related tweets across (left) and within (right) countries

topic significantly more than those at the extreme ends of the political spectrum (i.e., far-left or far-right parties and those with very high or very low GALTAN scores). For instance, members of the far-left GUE-NGL (The Left in the European Parliament) and the far-right ID (Identity and Democracy) mention Ukraine in roughly 9% and 10% of their tweets, respectively, while members of the liberal RE (Renew Europe) and the social democratic SD (Progressive Alliance of Socialists and Democrats) discuss Ukraine in over 20% and 15% of their post-invasion tweets, respectively.

6.6 Vader Sentiment Analysis

To assess the average sentiment in each of the 520,871 tweets, we use the VADER sentiment analyzer, as described above. In Fig. 6.6, we visualize the average daily sentiment of all tweets related to Ukraine and all tweets unrelated to Ukraine (as determined by the BERTopic model). Two observations are apparent: (1) the average sentiment in Ukraine-related tweets is consistently more negative than that in unrelated tweets. While Ukraine-related tweets are generally neutral or negative, unrelated tweets tend to be positive. This aligns with the sentiment hypothesis. (2) The divergence in sentiments between the two topics increases considerably after the Russian invasion of Ukraine. While the 95% confidence intervals for daily sentiments overlap

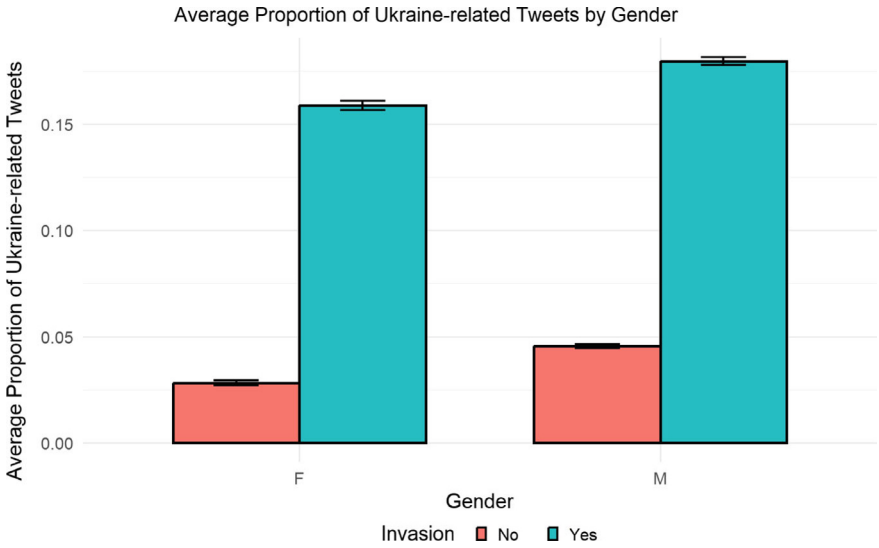

Fig. 6.3 Average proportion of Russia-Ukraine tweets before and after the invasion, split by gender. *Note* that the error bars represent 95% confidence intervals

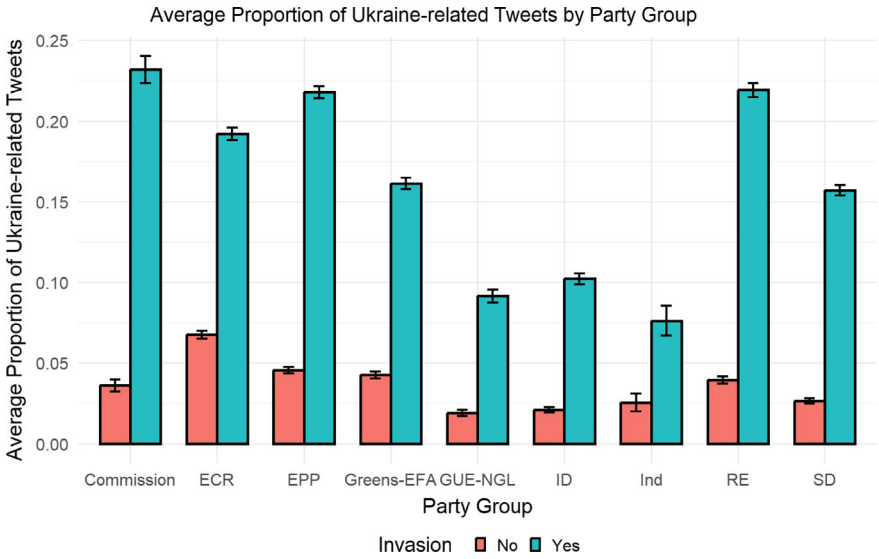

Fig. 6.4 Average proportion of Russia-Ukraine tweets before and after the invasion, split by party group. *Note* that the error bars represent 95% confidence intervals

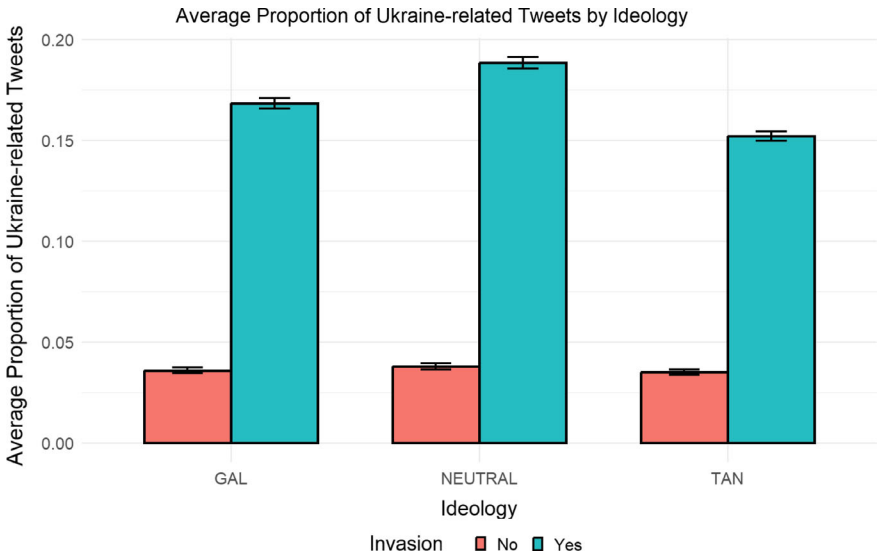

Fig. 6.5 Average proportion of Russia-Ukraine tweets before and after the invasion, split by GALTAN ideology. *Note* that the error bars represent 95% confidence intervals

to a large extent before the invasion, this is not the case for several months afterwards. Although the difference in sentiments between the two topics mainly was not statistically significant beforehand, it became significant following the invasion and remained so for a long period before gradually converging again roughly 100 days after February 24.

Having established that the Ukraine topic is discussed with significantly more negative sentiment than other topics, we next compare the sentiment of Ukraine-related tweets across the same strata analyzed with the topic model above: nationality, gender, party affiliation, and political ideology of EU politicians.

In the left panel of Fig. 6.7, we display the average sentiment of all Ukraine-related tweets within each country. Light colors indicate negative sentiments, dark colors represent positive sentiments. Apart from a few outliers, the sentiment in these tweets is overwhelmingly negative. However, contrary to the geographical sentiment hypothesis, it is not consistently more negative in countries closer to Ukraine. While some Eastern European countries are very negative (e.g., Estonia, Lithuania, or Poland), others are more positive (e.g., Latvia and Romania). Some Western European countries (e.g., the Netherlands and Germany) and some Southern European countries (e.g., Spain and Italy) are also very negative. The right panel of Fig. 6.7 shows the change in sentiment following the Russian invasion: for most countries, the political communication of their politicians shows a further increase in negative sentiments, with Scandinavian and Baltic countries experiencing particularly drastic

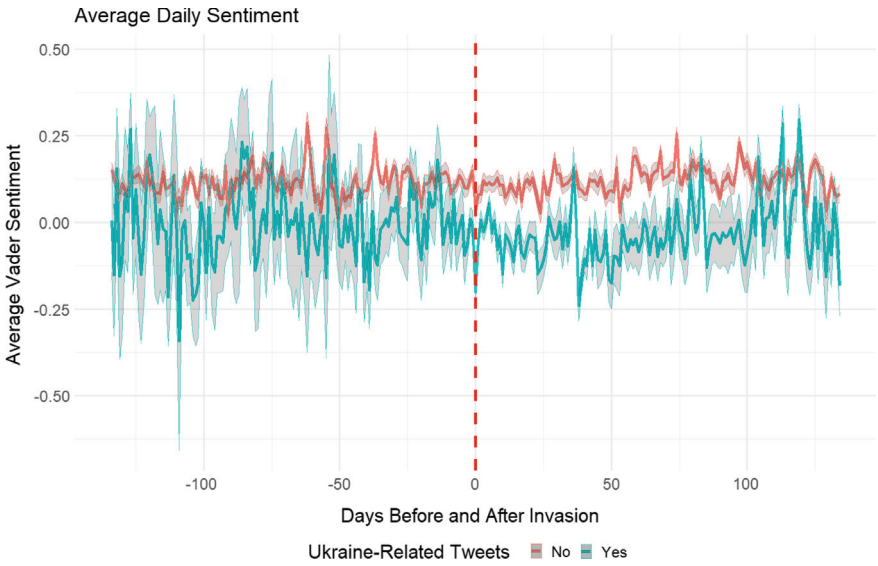

Fig. 6.6 Daily sentiment in Russia-Ukraine and non-related tweets. *Note* that the grey shades represent 95% confidence intervals

shifts in sentiment in the negative direction. Romanian politicians, who had previously held the most positive sentiments, also became significantly more negative after February 2022.

Further disaggregating the data, heterogeneity between sociopolitical groups becomes evident. In Fig. 6.8, we depict the average sentiment in Russia-Ukraine related tweets before and after the invasion, divided by gender. Male politicians are already somewhat more negative in their political communication than the female politicians prior to the invasion. After the invasion, the sentiment becomes significantly more negative for both genders, and it is significantly more negative for men than women. These findings confirm both the gendered sentiment hypothesis and the gendered polarization hypothesis.

Examining party affiliation in Fig. 6.9, the far-left GUE-NGL and the far-right ID are the most negative in their communication about Russia-Ukraine topics, while the more moderate RE and SD are more neutral, thus supporting the ideological sentiment hypothesis. Tweets from members of the EU Commission are significantly more positive than those from any parliamentary group. Across all parties and institutions, sentiment becomes more negative following the invasion, but, contrary to the ideological polarization hypothesis, this increase in negativity is not systematically stronger for ideologically extreme parties.

Finally, analyzing the political ideology of tweeting politicians (measured here by the GALTAN scale) in Fig. 6.10, the patterns of heterogeneity indicated by party affiliation remain salient. Politicians at both extremes of the political spectrum (i.e.,

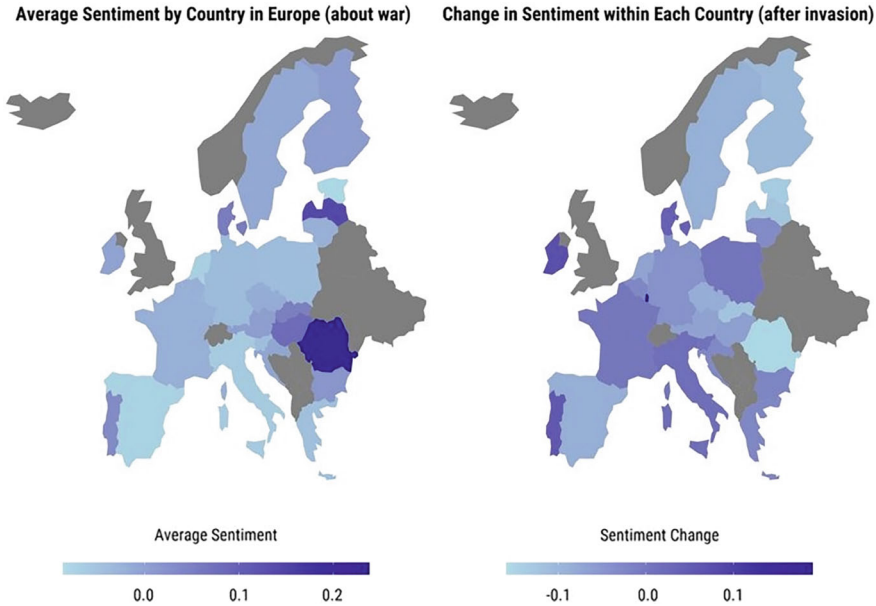

Fig. 6.7 Sentiment of all Russia-Ukraine related tweets (left) and sentiment-change after the invasion (right)

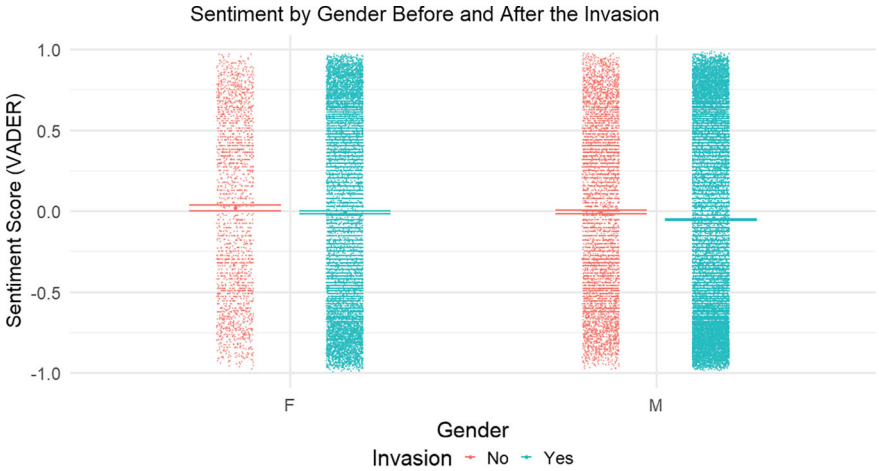

Fig. 6.8 Average sentiment in Russia-Ukraine tweets before and after the invasion, split by gender. *Note* that the error bars represent 95% confidence intervals

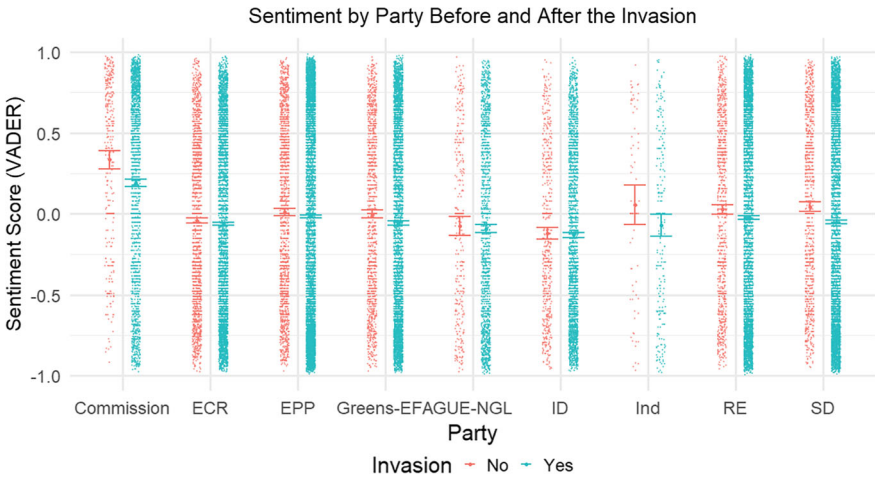

Fig. 6.9 Average sentiment in Russia-Ukraine tweets before and after the invasion, split by party affiliation. *Note* that the error bars represent 95% confidence intervals

those who are extremely green/alternative/liberal and those who are extremely traditional/authoritarian/nationalist) are more negative than moderate politicians, again supporting the ideological sentiment hypothesis.

Furthermore, unlike with party families, the gap in sentiment between ideologically extreme and moderate politicians on the GALTAN scale widens further after the invasion, supporting to the ideological polarization hypothesis.

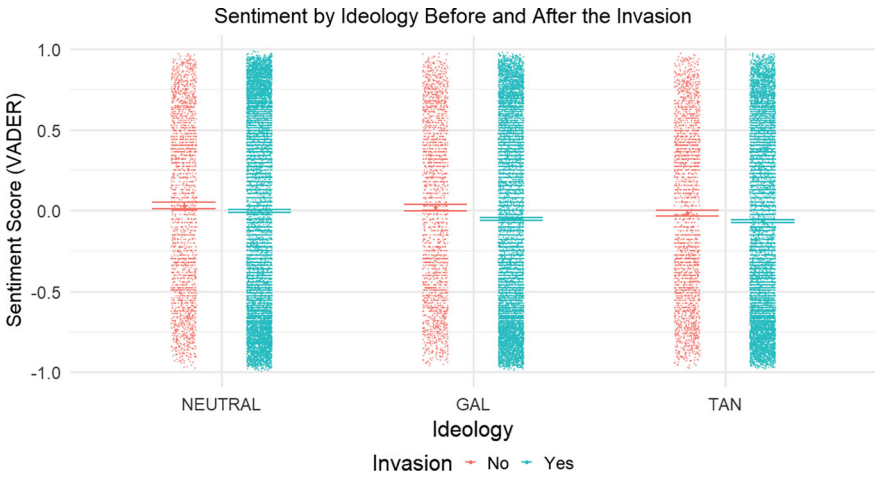

Fig. 6.10 Average sentiment in Russia-Ukraine tweets before and after the invasion, split by GALTAN ideology. *Note* that the error bars represent 95% confidence intervals

We also summarize our findings in Table 6.2, where we present the results from regression analyses (with standard errors clustered at the politician level). We show that (in line with H1) tweets after the invasion were 13 percentage points more likely to discuss the Russia-Ukraine topic ($p < 0.001$***), that (in line with H2a) tweets on the Russia-Ukraine topic are significantly more negative than others ($p < 0.001$***), that (in line with H2b) men tweeting about the war are significantly more negative than women ($p < 0.05$*), and that (in line with H2c) ideologically extreme (GAL or TAN) politicians are more negative than moderate politicians ($p < 0.05$* for both comparisons). The interaction coefficients for increasing sentiment divides between male/female and extreme/moderate politicians after the invasion are also (in line with H3) in the expected direction, but not statistically significant at the 5% level.

In sum, the sentiment and heterogeneity analyses reveal two trends: (1) communication about Russia-Ukraine topics became significantly more negative following the Russian invasion, much more so than communication about other topics, even though it started from a more negative baseline. (2) Those at the extreme ends of the political spectrum and men, who were already more negative than moderates

Table 6.2 Regression analyses for H1-H3

	Russia-Ukraine Topic (H1)	Sentiment (H2a)	Sentiment (H2b)	Sentiment (H2c)	Sentiment (H3)	Sentiment (H3)
(Intercept)	0.04*** (0.00)	0.12*** (0.01)	− 0.00 (0.01)	0.00 (0.02)	0.02 (0.02)	0.03 (0.02)
After Invasion	0.13*** (0.01)				− 0.03 (0.02)	− 0.03 (0.02)
Russia-Ukraine Topic		− 0.15*** (0.01)				
Male			− 0.04* (0.02)		− 0.03 (0.02)	
GAL				− 0.04* (0.02)		− 0.01 (0.01)
TAN				− 0.06** (0.02)		− 0.05 (0.03)
Male * After					−0.02 (0.02)	
GAL * After						−0.04 (0.03)
TAN * After						−0.02 (0.02)
Num. obs	512666	512666	55991	49298	55991	49298
R^2	0.05	0.01	0.00	0.00	0.00	0.00
Adj. R^2	0.05	0.01	0.00	0.00	0.00	0.00

***$p < 0.001$; **$p < 0.01$; *$p < 0.05$

Note Standard errors are clustered at the politician level

and women, began to grow more negative after the invasion, further widening the gap between moderate and radical politicians. These findings indicate an increase in polarization in political communication due to the Russian invasion of Ukraine.

However, when combining insights from the topic models with those from sentiment analyses, we see that those who are more moderate in their discussion of the Russia-Ukraine war (i.e., members of centrist parties and those with neutral GALTAN scores) tend to talk about the topic more on average. This shows that the more polarizing voices in the debate are not necessarily the dominant ones, except for the men. They are more negative in their Russia-Ukraine political communication before the invasion, discuss the topic more often than women, and become even more negative to a greater degree than women after the invasion.

6.7 Discussion and Conclusion

Do catastrophic events lead to polarization on social media? And do political elites exploit these sudden crises to boost their visability? To answer these questions, we analyzed the reactions of MEPs and EU Commissioners on Twitter/X to the Russian invasion of Ukraine on February 24, 2022. We compiled and machine-translated an original dataset of all MEPs and Commissioner tweets between October 13, 2021, and July 8, 2022 (Total N = 520.871), and enriched this data with a comprehensive set of individual-level covariates covering various political and sociodemographic groups.

Our hypotheses regarding attention-grabbing political communication during crises, sentiments expressed on social media, and the impact of a crisis on polarization among different social and political groups inform our design. For the attention hypothesis, we analyzed a BERTopic model to map the frequency of topics over time. The sentiment and polarization hypotheses were tested with the VADER sentiment analyzer, which we used to compare different sentiments between topics, within topics over time, and within topics across sociopolitical strata.

We found, consistent with our expectations and similar to others (Mainych et al. 2023), that politicians immediately concentrated their attention on the war in most Twitter posts, making this conflict the central issue of Spring 2022. In the weeks following the invasion (and even in the days before it), no other topic received nearly as much attention as the war in Ukraine.

Sentiment in conflict-related communication was consistently more negative than in other topics, and this difference in sentiment grew following the full-scale invasion. As hypothesized, male and ideologically extreme politicians expressed significantly more negative emotions than female and moderate politicians. However, contrary to our expectations, politicians from countries geographically closer to Ukraine were not systematically more negative than those from more distant countries. Spatial borders are becoming less meaningful for European politicians given the scope of Russia's military attack.

Polarization grew as the pre-war gap in sentiments widened between more negative men and extreme politicians on one side, and more positive women and moderate politicians on the other after the Russian invasion.

Since our findings are based on a single case, their scope is limited. Future research should expand the database to include different events, various politician samples, and multiple social media platforms, ideally combining observational and (quasi-) experimental methods to establish causality and external validity.

Still, our results provide compelling empirical evidence that European political elites are emotionally triggered by external shocks like the Russia-Ukraine war. They utilize negative emotions especially at the end of the ideological spectrum, thereby activating extreme views and contributing to polarization on social networks. While emotions directly influence behavior, we also show that they do not last long. After 100 days of war, politicians have mostly returned to normal. Thus, political polarization is a volatile and malleable phenomenon (Axelrod et al. 2021; Macy et al. 2021). However, in times of poly-crises (Tooze 2022), constantly new events may sustain the exceptional emotional state.

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